ANUSHKA GUPTA

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PROFESSIONAL SUMMARY

High-achieving M.Tech (AI & ML) student at SRM Institute of Science and Technology with a CGPA of 9.445/10. Accomplished researcher with a proven track record in Healthcare AI, Computer Vision, and Deep Learning, featuring peer-reviewed publications in Springer and CRC Press. Expertise spans across Machine Learning, Full Stack Development, and Cyber Security, complemented by hands-on experience in building scalable AI-driven solutions and intelligent surveillance systems.

RESEARCH INTERESTS

Artificial Intelligence • Machine Learning • Deep Learning • Computer Vision • Healthcare AI • Graph Neural Networks (GNNs) • Explainable AI (XAI)

EDUCATION

SRM Institute of Science and Technology| 2025 – 2027

M.Tech in Computer Science & Engineering (AI & ML)|**CGPA: 9.445/10**

KIET Group of Institutions| 2021 – 2025

B.Tech in Information Technology|**CGPA: 8.11/10**

PUBLICATIONS & PATENTS

- **Publication:** Gupta, A., Mehrotra, S., Kumar, H., & Nagappan, P. (2025). "Early Prediction and Diagnosis of Arthritis: A Novel Multimodal Hybrid Ensemble AI Approach."*Springer Nature Switzerland*, pp. 392–401.
- **Publication:** Kumar, V., Singh, A. P., Gupta, A., Sharma, A., & Vashisht, R. (2025). "A Comparative Analysis of Face Recognition Models Using Deep Learning."*CRC Press*, pp. 761–767.
- **Patent:** Contributor to "AI-Based Face Recognition and Missing Person Identification System."

PROFESSIONAL EXPERIENCE

FeathrTech, Noida| Full Stack Developer Intern | May 2025 – Jul 2025

- Architected a full-scale EdTech platform using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Engineered secure authentication systems, dynamic course management modules, and optimized RESTful APIs.
- Enhanced backend performance and user interface responsiveness, resulting in improved platform usability.

Coding Ninjas| Data Science Trainee Intern

- Executed end-to-end data pipelines including exploratory data analysis (EDA), feature engineering, and predictive modeling.
- Implemented robust machine learning algorithms using Python and Scikit-Learn to solve complex analytical problems.

Cisco| Cyber Security Intern

- Performed comprehensive vulnerability assessments and network packet analysis to secure digital assets.
- Identified critical security loopholes and formulated strategic mitigation plans to enhance system integrity.

TECHNICAL PROJECTS

- **Cognitive Trajectory-Aware Graph Transformer (CTGT):**Engineered a multimodal Graph Transformer model integrating MRI data and longitudinal cognitive assessments for the early detection of Alzheimer's Disease.
- **Deep Learning-Based Face Recognition Analysis (CRC Press, 2025):**Conducted a comparative study of DeepFace, InsightFace, and OpenFace models for biometric authentication and face recognition. Evaluated model performance using Precision, Recall, and F1-Score metrics,leading to published research and patent contributions.
- **Dynamic Multimodal Hybrid Ensemble for Arthritis Prediction (Springer, 2025):**Developed a multimodal AI framework integrating clinical, imaging, biomarker, and wearable sensor data for early arthritis prediction. Combined LightGBM, ResNet-50, and LSTM models with attention-based fusion to improve predictive performance.

- **UCF-Net: Hybrid CNN–Transformer Framework for Cervical Cancer Classification:** Built a ResNet50–ViT–U-Net model achieving 95.0% accuracy and 0.9960 ROC-AUC on the SIPaKMeD dataset with Explainable AI (Grad-CAM). •
- **Federated Multi-Task Deep Neural Network for Knee Osteoarthritis Prediction :** Developed a federated learning framework using gait and clinical data, achieving 93.47% diagnosis accuracy with SHAP-based explainability.

TECHNICAL SKILLS

- **Languages:**Python, SQL, JavaScript, HTML, CSS
- **AI/ML:**Deep Learning, Computer Vision, NLP, Generative AI, Graph Neural Networks
- **Frameworks:**TensorFlow, PyTorch, Scikit-Learn, OpenCV, React.js, Node.js
- **Tools:**Git, GitHub, Jupyter Notebook, Google Colab, Vs Code

CERTIFICATIONS & ACHIEVEMENTS

- **Academic Excellence:**Maintained a 9.445 CGPA in M.Tech (Year 1).
- **Certifications:**AWS Cloud Foundations, Data Science (Coding Ninjas), Cyber Security (Cisco virtual).
- **Impact:**Authored two international research papers and contributed to a pending AI patent.